PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

Construction Site Management System

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Project Guide
Ms. Avani Nair

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- Objectives
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1. Introduction

- **Problem Identified :**

- Construction site operations are managed manually using paper registers and messaging apps.

- Poor coordination, data inconsistency, and lack of real-time visibility in attendance, progress, and material usage.

- **Solution Proposed :**

- A centralized digital Construction Site Management System.

- A web-based platform to manage workforce, materials, progress tracking, and reporting efficiently.

2. Objectives

1. To store and manage construction site data including worker details, attendance records, project tasks, and material information in a centralized system.

2. To streamline material requests, approvals, and usage tracking to reduce delays and resource mismanagement.

3. To generate essential operational reports such as attendance summaries, daily progress reports, material usage, and billing records to support effective project monitoring and decision-making.

4. To track workforce attendance and daily site progress accurately using digital reporting and location-based validation.

3. Scope

1. Can be applied in small, medium, and large construction projects.
2. Can be used by site managers, contractors, engineers, and supervisors.
3. Can be deployed across multiple construction sites.
4. Can be extended for enterprise-level construction management.
5. Can be integrated with billing and compliance systems in future.

4. Feature /Functionality

Feature 1: Workforce Management
Maintains worker details, attendance records, and role assignments. Enables location-based attendance validation.

Feature 2: Daily Progress Tracking
Allows supervisors to update daily work progress with structured digital reports and updates.

Feature 3: Material Management
Tracks material requests, approvals, usage, and stock availability to prevent shortages and delays.

Feature 4: Reporting & Dashboard
Provides interactive dashboards for attendance summaries, material usage, cost tracking, and project progress.

Feature 5: Real-Time Synchronization
Instant updates across devices ensuring remote stakeholders can monitor site activity.

1. Sustainable Development Goal 8 : Decent Work and Economic Growth

Why it fits:

Tracks worker attendance and ensures **fair & timely wages**

Improves **working conditions and transparency**

Reduces exploitation and mismanagement of labor

2. Sustainable Development Goal 9 : Industry, Innovation and Infrastructure

Why it fits:

Introduces a **digital platform** in a traditionally manual industry

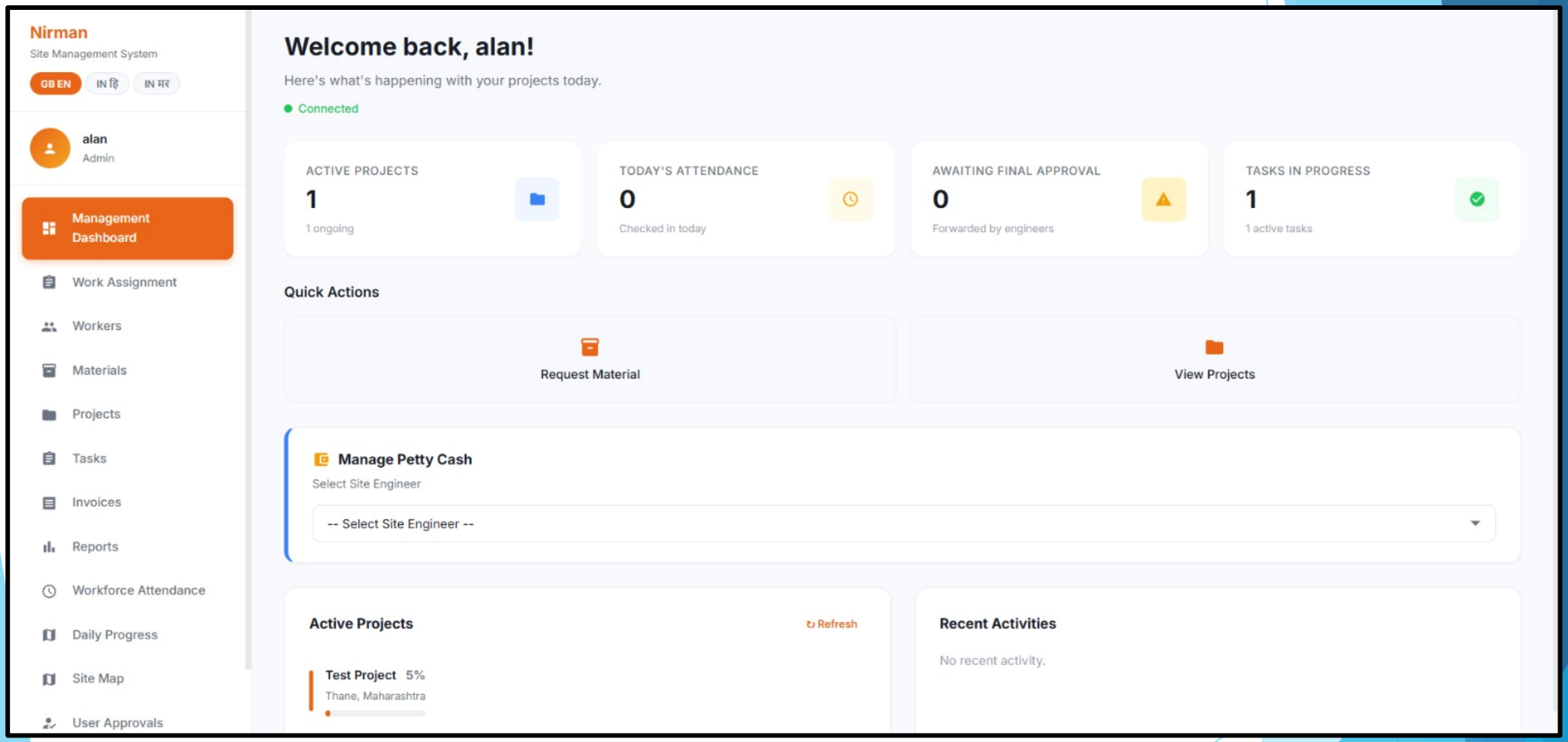
Enhances **efficiency in construction operations**

Supports development of **modern, sustainable infrastructure.**

5. Outcome of Project

1. User can log in securely.
2. User can add and manage worker details.
3. User can mark and verify attendance digitally.
4. User can update daily site progress.
5. User can request and approve materials.
6. User can generate attendance and progress reports.
7. Admin can monitor overall project performance through dashboard.

FRONTEND


Nirman
Site Management System
GB EN
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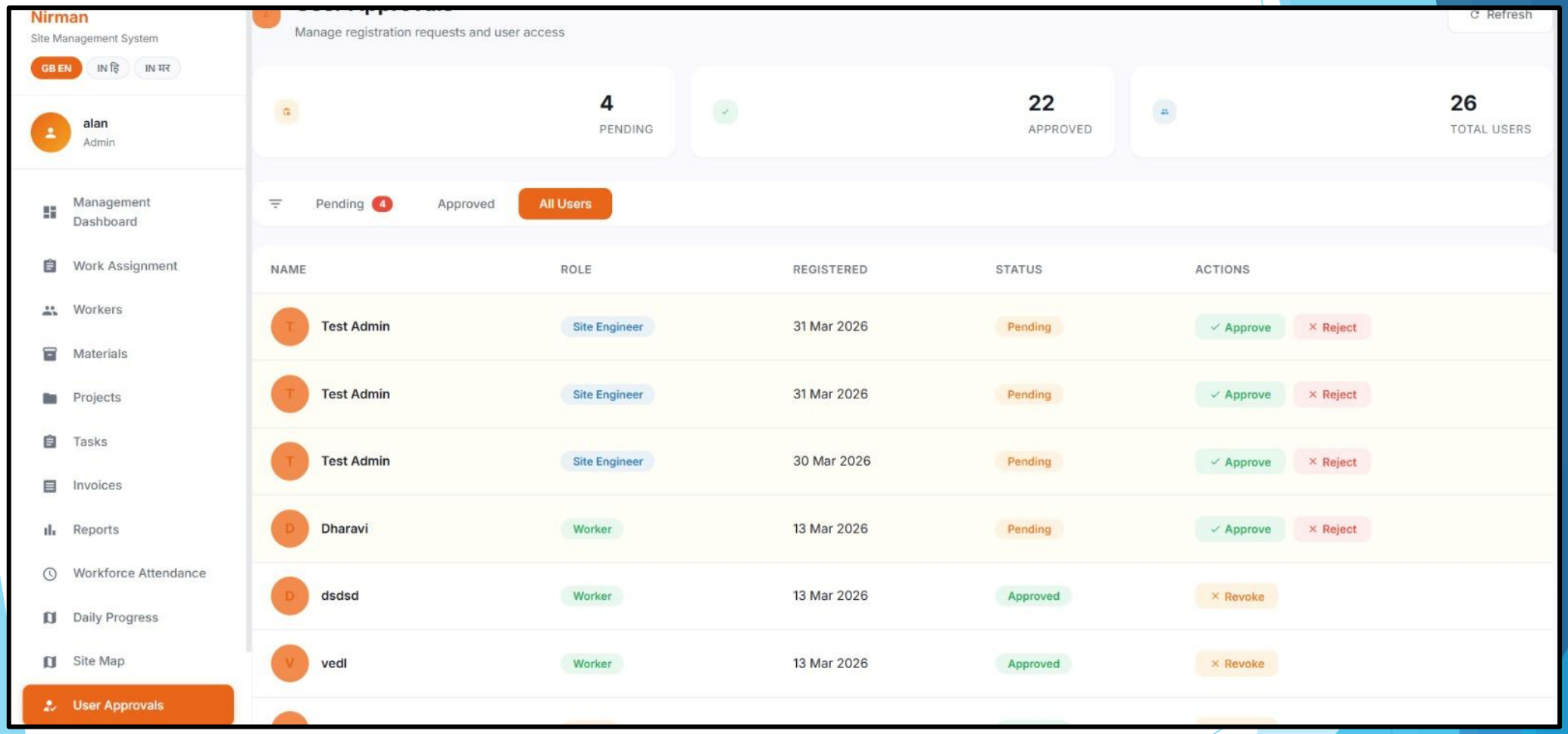
alan
Admin
Management Dashboard
Work Assignment
Workers
Materials
Projects
Tasks
Invoices
Reports
Workforce Attendance
Daily Progress
Site Map
User Approvals
Welcome back, alan!
Here's what's happening with your projects today.
Connected
ACTIVE PROJECTS
1
1 ongoing
TODAY'S ATTENDANCE
0
Checked in today
AWAITING FINAL APPROVAL
0
Forwarded by engineers
TASKS IN PROGRESS
1
1 active tasks
Quick Actions
Request Material
View Projects
Manage Petty Cash
Select Site Engineer
-- Select Site Engineer --
Active Projects
Refresh
Test Project 5%
Thane, Maharashtra
Recent Activities
No recent activity.

FRONTEND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Site Management System
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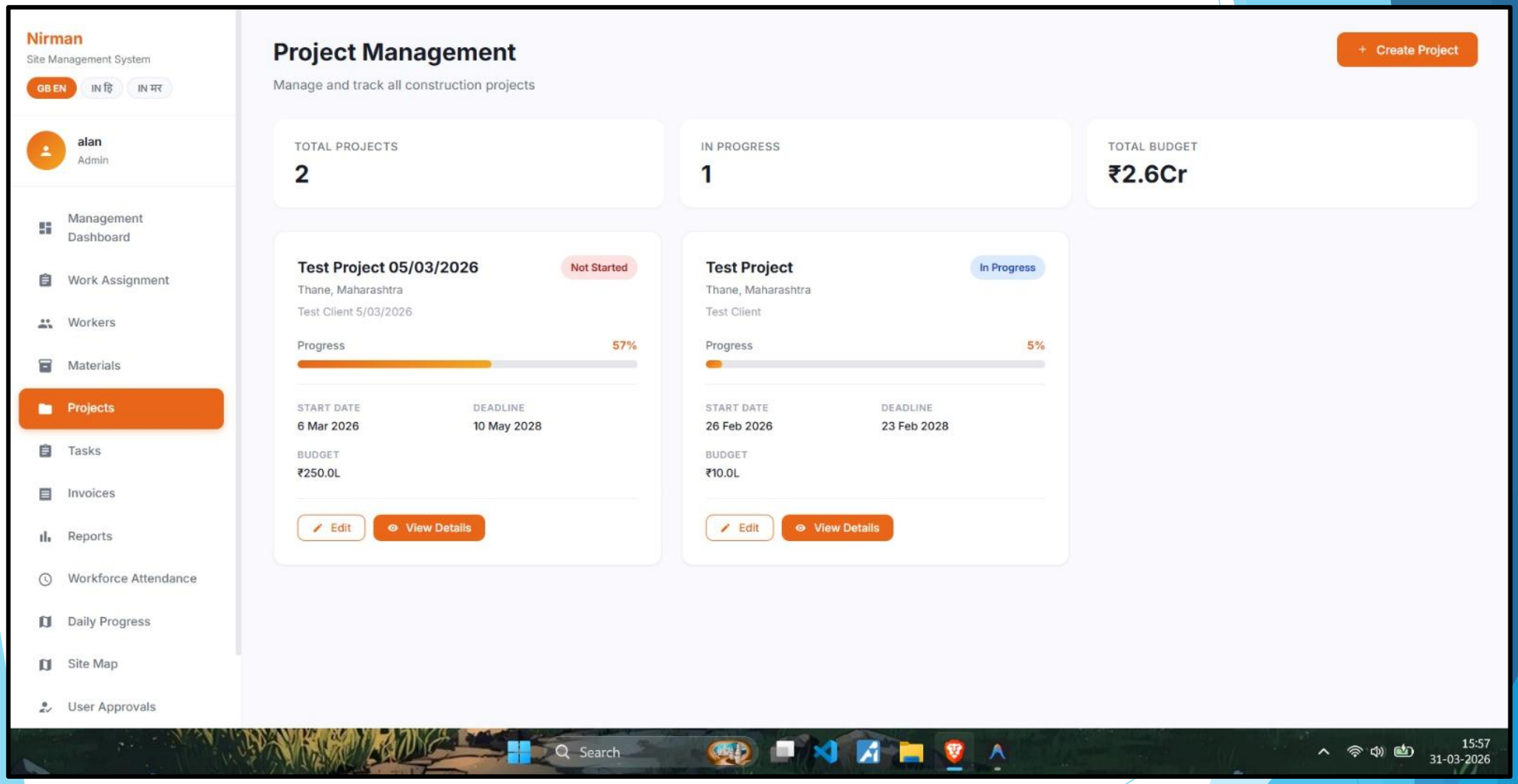
IN मर
alan
Admin
Management Dashboard
Work Assignment
Workers
Materials
Projects
Tasks
Invoices
Reports
Workforce Attendance
Daily Progress
Site Map
User Approvals
Manage registration requests and user access
Refresh
4
PENDING
22
APPROVED
26
TOTAL USERS
Pending 4
Approved
All Users
NAME
ROLE
REGISTERED
STATUS
ACTIONS
Test Admin
Site Engineer
31 Mar 2026
Pending
Approve
Reject
Test Admin
Site Engineer
31 Mar 2026
Pending
Approve
Reject
Test Admin
Site Engineer
30 Mar 2026
Pending
Approve
Reject
Dharavi
Worker
13 Mar 2026
Pending
Approve
Reject
dsdsd
Worker
13 Mar 2026
Approved
Revoke
vedl
Worker
13 Mar 2026
Approved
Revoke

FRONTEND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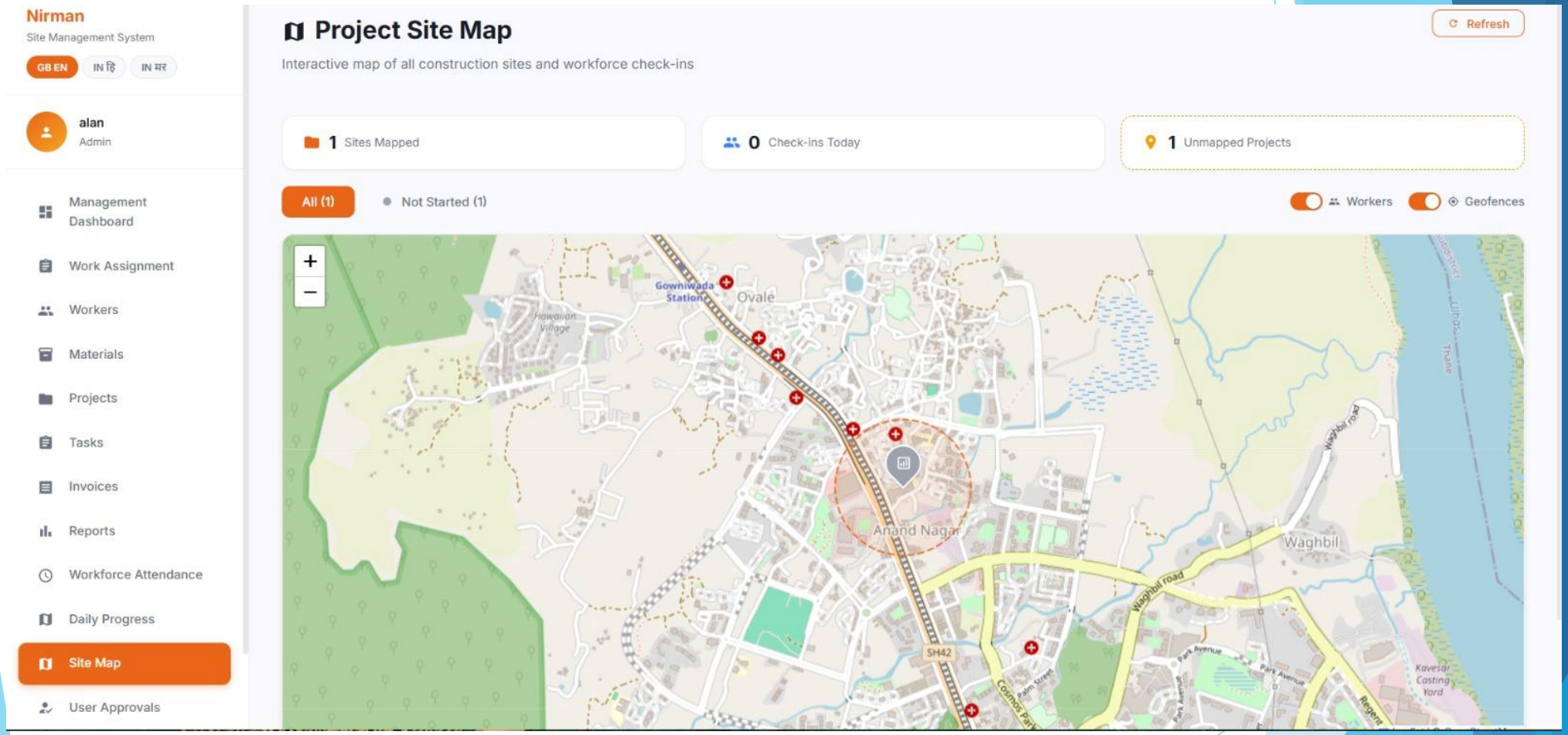
Admin
Management Dashboard
Work Assignment
Workers
Materials
Projects
Tasks
Invoices
Reports
Workforce Attendance
Daily Progress
Site Map
User Approvals
Project Management
Manage and track all construction projects
+ Create Project
TOTAL PROJECTS
2
IN PROGRESS
1
TOTAL BUDGET
₹2.6Cr
Test Project 05/03/2026
Not Started
Thane, Maharashtra
Test Client 5/03/2026
Progress
57%
START DATE
6 Mar 2026
DEADLINE
10 May 2028
BUDGET
₹250.0L
Edit
View Details
Test Project
In Progress
Thane, Maharashtra
Test Client
Progress
5%
START DATE
26 Feb 2026
DEADLINE
23 Feb 2028
BUDGET
₹10.0L
Edit
View Details
Search
15:57
31-03-2026

FRONTEND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Site Management System
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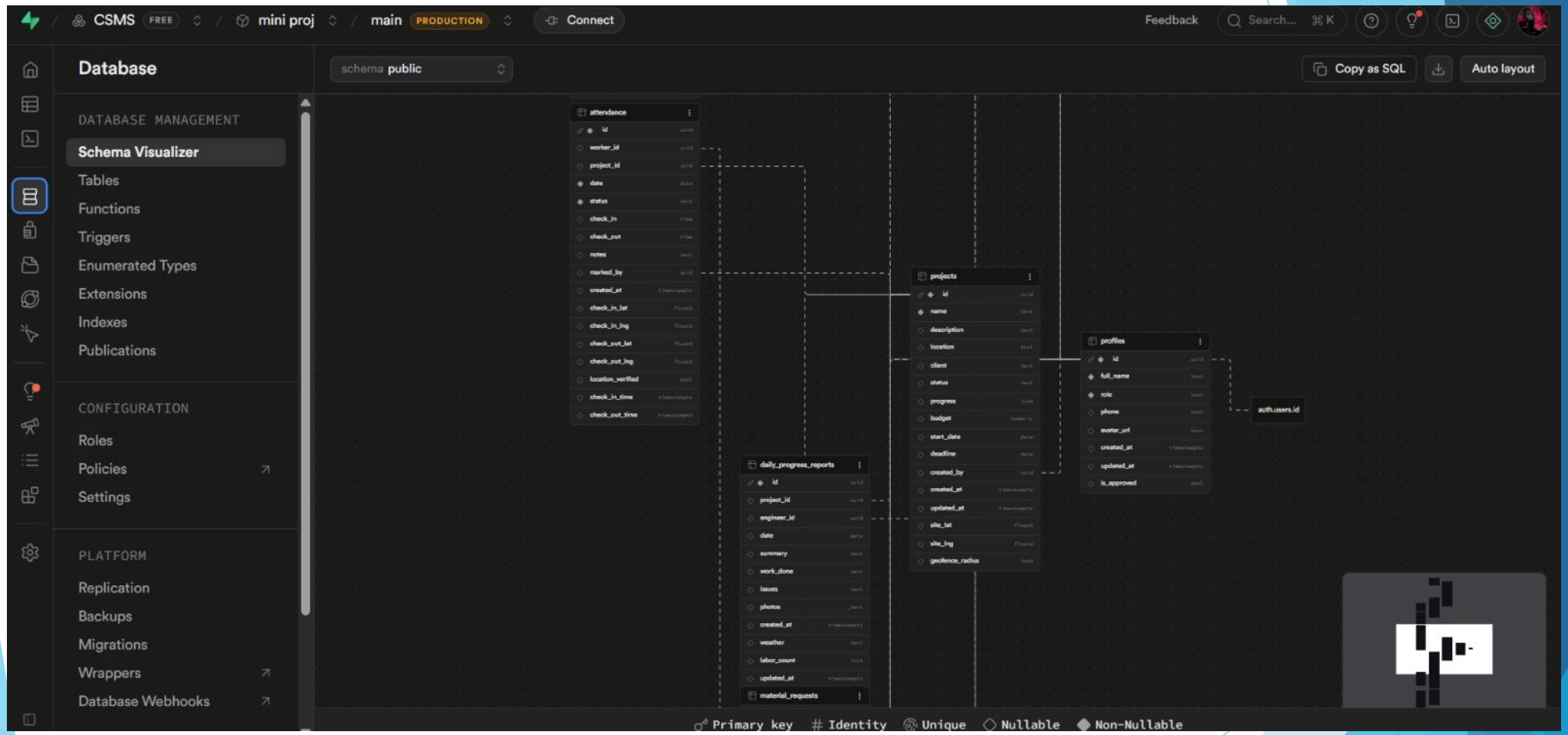
IN मर
alan
Admin
Management Dashboard
Work Assignment
Workers
Materials
Projects
Tasks
Invoices
Reports
Workforce Attendance
Daily Progress
Site Map
User Approvals
Project Site Map
Refresh
Interactive map of all construction sites and workforce check-ins
1 Sites Mapped
0 Check-ins Today
1 Unmapped Projects
All (1)
Not Started (1)
Workers
Geofences
Gowniwada Station
Ovale
Anand Nagar
Waghbil
Waghbil road
SH42
Park Avenue
Palm Street
Cosmos Park
Kavesar Casting Yard
Regent

BACKEND


CSMS FREE
mini proj
main PRODUCTION
Connect
Feedback
Search... ⌘K
Database
schema public
Copy as SQL
Auto layout
DATABASE MANAGEMENT
Schema Visualizer
Tables
Functions
Triggers
Enumerated Types
Extensions
Indexes
Publications
CONFIGURATION
Roles
Policies
Settings
PLATFORM
Replication
Backups
Migrations
Wrappers
Database Webhooks
attendance
projects
profiles
daily_progress_reports
material_requests
auth.users.id
Primary key
Identity
Unique
Nullable
Non-Nullable

6. Technology Stack

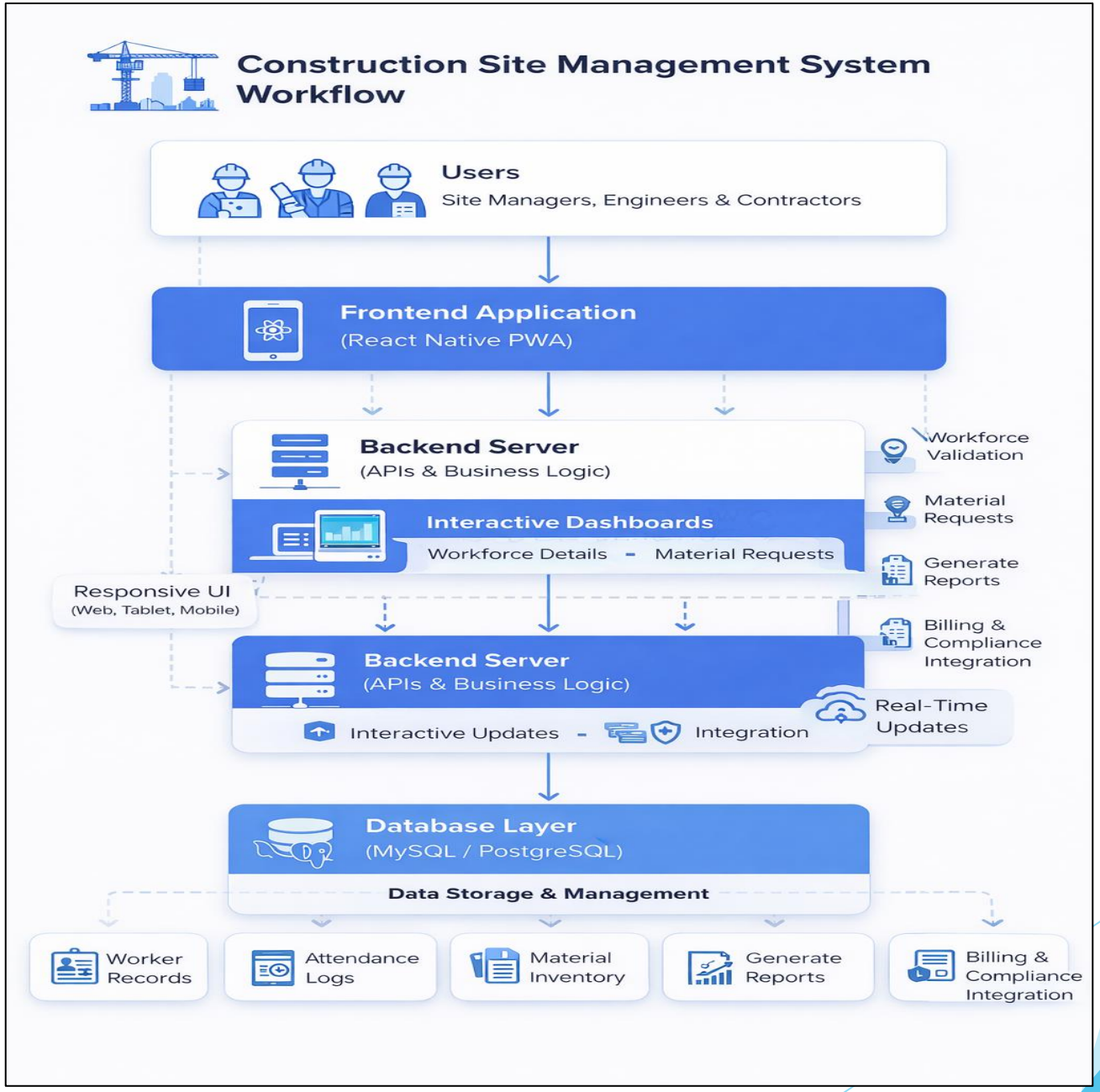
- **Front-End (GUI):**

1. React Native (Progressive Web App – PWA)
2. Responsive UI for web, tablet, and mobile devices

- **Back-End:**

1. MySQL / PostgreSQL (Relational Database)
2. Centralized data storage and management
3. Supabase

7. Block Diagram



Thank You...!!